

A Physics-Informed Neural Surrogate for Stellar Chemical Equilibrium in `jorg`

Implementation Design Note

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Abstract

This document specifies the current PINN design for fast chemical-equilibrium inference in `jorg`. The reference physics follows the Korg formulation implemented in `Korg.jl/src/statmech.jl` and ported in `jorg/src/jorg/statmech/korg_chemical_equilibrium.py`. The PINN predicts electron density and per-element neutral fractions from thermodynamic state and abundances, and is trained with a two-stage objective: (i) supervised regression to trusted solver outputs, and (ii) residual minimization of atomic element conservation and charge neutrality. The present physics loss is atomic-only; molecular constraints are delegated to the label generator and not yet enforced explicitly in the PINN objective.

1 Problem Statement and Scope

Given

$$(T, n_{\text{tot}}, \mathbf{A}), \quad \mathbf{A} = (A_1, \dots, A_{92}), \quad \sum_{Z=1}^{92} A_Z = 1, \quad (1)$$

we seek a fast approximation to the equilibrium solution

$$(n_e, \mathbf{f}), \quad \mathbf{f} = (f_1, \dots, f_{92}), \quad (2)$$

where n_e is electron number density and f_Z is the neutral fraction of element Z .

The reference equilibrium labels are produced by the trusted backends:

- `jorg/src/jorg/statmech/chem_eq_jax.py` (default),
- `jorg/src/jorg/statmech/korg_chemical_equilibrium.py` (SciPy port of Korg).

The current PINN loss enforces only atomic constraints (neutral, singly ionized, doubly ionized species), consistent with the present phase of development.

2 Reference Physics: Atomic Ionization Structure

2.1 Saha ionization weights

For each element Z , define Saha ratios

$$w_{Z,\text{II}} = \frac{n_{Z,\text{II}}}{n_{Z,\text{I}}}, \quad (3)$$

$$w_{Z,\text{III}} = \frac{n_{Z,\text{III}}}{n_{Z,\text{I}}}. \quad (4)$$

Following `statmech.jl`, the first-ionization ratio is

$$w_{Z,\text{II}} = \frac{2}{n_e} \frac{U_{Z,\text{II}}(T)}{U_{Z,\text{I}}(T)} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \exp\left(-\frac{\chi_{Z,1}}{k_B T}\right), \quad (5)$$

and the second-ionization ratio is

$$w_{Z,\text{III}} = w_{Z,\text{II}} \frac{2}{n_e} \frac{U_{Z,\text{III}}(T)}{U_{Z,\text{II}}(T)} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \exp\left(-\frac{\chi_{Z,2}}{k_B T}\right), \quad (6)$$

with the hydrogen special case $w_{1,\text{III}} = 0$ in the implementation.

2.2 Atomic number densities

Define the nucleus reservoir for element Z as

$$n_{\text{nuc},Z} = A_Z (n_{\text{tot}} - n_e). \quad (7)$$

Given neutral fraction f_Z , the three atomic stages used in the PINN loss are

$$n_{Z,\text{I}} = f_Z n_{\text{nuc},Z}, \quad (8)$$

$$n_{Z,\text{II}} = w_{Z,\text{II}} n_{Z,\text{I}}, \quad (9)$$

$$n_{Z,\text{III}} = w_{Z,\text{III}} n_{Z,\text{I}}. \quad (10)$$

3 PINN Parameterization

3.1 Inputs and outputs

The model input is the 94-dimensional vector

$$\mathbf{x} = [\log_{10} T, \log_{10} n_{\text{tot}}, \mathbf{A}]. \quad (11)$$

The network output has 93 channels:

$$(\mathbf{z}_f, \mathbf{z}_e), \quad \mathbf{z}_f \in \mathbb{R}^{92}, \mathbf{z}_e \in \mathbb{R}. \quad (12)$$

It is mapped to physical variables as

$$\hat{f}_Z = \text{clip}(\sigma(\mathbf{z}_{f,Z}), 10^{-12}, 1), \quad (13)$$

$$\hat{n}_e = n_{\text{tot}} \sigma(\mathbf{z}_e), \quad (14)$$

which guarantees $0 < \hat{f}_Z \leq 1$ and $0 < \hat{n}_e < n_{\text{tot}}$.

3.2 Architecture

The baseline architecture in `chem_eq.pinn.models.py` is an MLP with hidden widths

$$(256, 512, 512, 256, 128) \quad (15)$$

and Swish activation. A Flax implementation is used when available, with a pure-JAX fallback path preserving the same functional mapping.

4 Training Objectives

4.1 Supervised objective

Training labels $(n_e^*, \mathbf{f}^*)$ are generated from the reference solvers. The supervised objective is

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{n_e} + \mathcal{L}_f, \quad (16)$$

$$\mathcal{L}_{n_e} = \left\langle [\log_{10} \hat{n}_e - \log_{10} n_e^*]^2 \right\rangle, \quad (17)$$

$$\mathcal{L}_f = \left\langle [\log_{10} \hat{f}_Z - \log_{10} f_Z^*]^2 \right\rangle, \quad (18)$$

with numerical floors applied in code before logarithms.

4.2 Physics objective

The PINN physics residuals are computed from $\hat{n}_e$ and $\hat{\mathbf{f}}$.

Element conservation residual

$$r_{\text{elem},Z} = \frac{n_{Z,\text{I}} + n_{Z,\text{II}} + n_{Z,\text{III}} - n_{\text{nuc},Z}}{\max(n_{\text{nuc},Z}, \epsilon)}, \quad \epsilon = 10^{-100}. \quad (19)$$

The loss uses abundance-weighted MSE:

$$\mathcal{L}_{\text{elem}} = \left\langle \frac{1}{92} \sum_{Z=1}^{92} \omega_Z r_{\text{elem},Z}^2 \right\rangle, \quad \omega_Z = \frac{\max(A_Z, 10^{-12})}{\langle \max(A_Z, 10^{-12}) \rangle_Z}. \quad (20)$$

Charge-neutrality residual

$$r_{\text{chg}} = \frac{\sum_{Z=1}^{92} (n_{Z,\text{II}} + 2n_{Z,\text{III}}) - n_e}{\max(n_e, \epsilon)}, \quad \mathcal{L}_{\text{chg}} = \langle r_{\text{chg}}^2 \rangle. \quad (21)$$

The total physics loss is

$$\mathcal{L}_{\text{phys}} = w_{\text{elem}} \mathcal{L}_{\text{elem}} + w_{\text{chg}} \mathcal{L}_{\text{chg}}, \quad (22)$$

with current defaults $w_{\text{elem}} = 1$, $w_{\text{chg}} = 10$.

4.3 Two-phase optimization

The default schedule is:

1. supervised pretraining with $\mathcal{L}_{\text{sup}}$;
2. phase-2 fine tuning with either $\mathcal{L}_{\text{phys}}$ only or

$$\mathcal{L}_{\text{phase2}} = w_{\text{sup},2} \mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{phys}}. \quad (23)$$

Optimization uses AdamW with warmup and cosine decay (`chem_eq_pinn_train.py`).

5 Data Generation and Solver Coupling

Training/validation tuples are

$$(T, n_{\text{tot}}, \mathbf{A}, n_e^*, \mathbf{f}^*), \quad (24)$$

sampled over broad stellar-atmosphere ranges in $(T, \log_{10} n_{\text{tot}}, [M/H], [\alpha/H])$ and filtered for physically valid solutions.

Two points are important for interpretation:

- The reference solvers may include molecular equilibrium terms, depending on supplied equilibrium-constant tables.
- The current PINN physics loss does not yet include explicit molecular residual equations.

Therefore, molecular effects are presently learned through supervision rather than enforced as hard residual constraints.

6 Software Mapping

File	Functional role
<code>chem_eq_pinn_models.py</code>	Network architecture and output parameterization
<code>chem_eq_pinn_loss.py</code>	Saha weights, residuals, and physics losses
<code>chem_eq_pinn_train.py</code>	Data generation, objectives, and optimization loops
<code>chem_eq_pinn.py</code>	High-level train/infer/save/load API
<code>korg_chemical_equilibrium.py</code>	Full SciPy reference solver (Korg port)
<code>Korg.jl/src/statmech.jl</code>	Upstream physical formulation and algorithmic reference

7 Validation Protocol

Model quality is assessed with:

- supervised agreement metrics (e.g. mean and percentile relative errors in n_e and f_Z),
- physics diagnostics ($|r_{\text{elem},Z}|$ and $|r_{\text{chg}}|$ distributions),
- out-of-sample checks over temperature, density, and metallicity slices.

For operational reproducibility, cached datasets are recommended via `examples/train_chem_eq_pinn_full.py` options:

- `--prepare-data-only`,
- `--load-data`,
- `--data-dir`.

8 Current Limitations and Next Steps

- Physics regularization is atomic-only; molecular equilibrium constraints are not yet explicit in $\mathcal{L}_{\text{phys}}$.
- The surrogate is calibrated over a finite thermodynamic domain; extrapolation to extreme conditions requires dedicated stress testing.
- A direct extension path is to augment the residual system with molecular conservation equations consistent with the Korg formalism.